Langevin Coupling of Oriented Disks

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1 Rotational Langevin equation for a 2D hexagonal disk array

We consider a two-dimensional triangular lattice of identical rigid disks of radius a. The disk centres are assumed to be fixed on the lattice, and each disk j can only rotate about its own centre. The corresponding angular coordinate and angular velocity are

$$\theta_j = \theta_j(t), \qquad \omega_j = \dot{\theta}_j. \tag{1.1}$$

The nearest-neighbour lattice spacing is $\ell = 2a + h_0$, where h_0 is the mean gap width between neighbouring disks. We work in the lubrication limit,$h_0 \ll a$, and assume creeping flow, so that the fluid in the thin gaps is described by Stokes flow and the lubrication approximation.

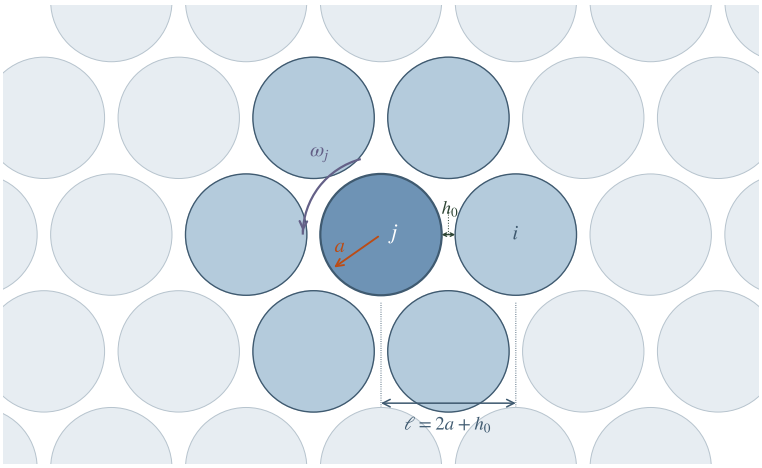


Figure 1: Schematic of the two-dimensional triangular lattice of rigid disks. The central disk j and its six nearest neighbours are highlighted. The lattice spacing is $\ell = 2a + h_0$, where a is the disk radius and h_0 is the mean gap width. Each disk rotates about its own centre with angular velocity ω_j.

1.1 Lubrication torque between two disks

Consider two neighbouring disks i and j. Let Γ_{ij} denote the thin-gap contact region between them, and let s be a coordinate along this gap. The local gap height is $h_{ij}(s)$. At a point on the gap, the surface velocity of disk j due to rotation is $\mathbf{u}_j(\mathbf{x}) = \omega_j\,\hat{\mathbf{z}} \times (\mathbf{x} - \mathbf{r}_j)$. For equal disks and nearest-neighbour geometry, the relative tangential surface velocity across the gap is proportional to the *sum* of the two angular velocities,

$$\Delta u_{ij}(s) \simeq a\big(\omega_i + \omega_j\big), \tag{1.2}$$

because the two facing surface points lie on the same side of their respective centres and their velocities add: for a co-rotating pair $\omega_i = \omega_j = \omega$ the facing surfaces slide past one another at $2a\omega$, whereas a counter-rotating pair would roll without slipping. In the lubrication limit, the shear stress transmitted through the gap is of the form

$$\tau_{ij}(s) \simeq \mu \frac{\Delta u_{ij}(s)}{h_{ij}(s)} = \mu a \frac{\omega_i + \omega_j}{h_{ij}(s)}, \tag{1.3}$$

where μ is the dynamic viscosity. The resulting hydrodynamic torque on disk j from the gap with disk i is obtained by integrating the shear stress over the lubrication region,

$$T_{ij}^{\mathrm{lub}} = -a \int_{\Gamma_{ij}} \tau_{ij}(s)\,\mathrm{d}s = -\mu a^2(\omega_i + \omega_j) \int_{\Gamma_{ij}} \frac{\mathrm{d}s}{h_{ij}(s)}. \tag{1.4}$$

It is convenient to define Γ_{ij} the dimensionless-in-time coupling coefficient

$$T_{ij}^{\mathrm{lub}} = -I\,\Gamma_{ij}\,(\omega_i + \omega_j), \qquad \Gamma_{ij} = \frac{\mu a^2}{I}\int_{\Gamma_{ij}} \frac{\mathrm{d}s}{h_{ij}(s)}, \tag{1.5}$$

where I is the moment of inertia of one disk per unit thickness. For a uniform hexagonal lattice, all six nearest-neighbour gaps are equivalent, so $\Gamma_{ij} = \Gamma$ for all nearest neighbours i, j .

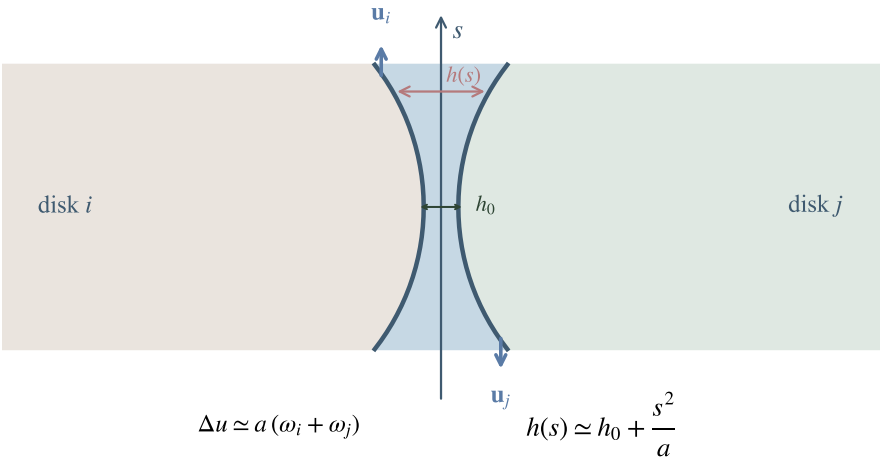


Figure 2: Enlarged view of the lubrication gap between two neighbouring disks i and j. The local gap height is $h(s) \simeq h_0 + s^2/a$. The relative tangential surface velocity is $\Delta u \simeq a(\omega_i + \omega_j)$, and the shear stress in the gap is $\tau \simeq \mu\Delta u/h(s)$. The integration region Γ_{ij} along the gap is indicated.

1.2 The effective Langevin equation

To begin with, the angular momentum balance for disk j is

$$I\dot{\omega}_j = -\gamma_{\mathrm{bulk}}\omega_j + \sum_{i\in\mathcal{N}(j)} T_{ij}^{\mathrm{lub}} + T_j^{\mathrm{st}}(t), \tag{1.6}$$

where γ_{bulk} is the (bulk) rotational damping from the surrounding fluid, $\mathcal{N}(j)$ denotes the six nearest neighbours of j, and $T_j^{\mathrm{st}}(t)$ is the stochastic torque from thermal fluid fluctuations. Substituting the lubrication torque for the six nearest neighbours gives

$$\dot{\omega}_j = -\frac{\gamma_{\mathrm{bulk}}}{I}\omega_j - \sum_{i\in\mathcal{N}(j)} \Gamma_{ij}(\omega_j + \omega_i) + \frac{1}{I}T_j^{\mathrm{st}}(t) = -\left(\frac{\gamma_{\mathrm{bulk}}}{I} + 6\Gamma\right)\omega_j - \Gamma\sum_{\langle i,j\rangle}\omega_i + \frac{1}{I}T_j^{\mathrm{st}}(t), \tag{1.7}$$

where $\sum_{\langle i,j\rangle}\omega_i$ denotes the sum over the six nearest neighbours of j. Rearranging,

$$\dot{\omega}_j + \left(\frac{\gamma_{\mathrm{bulk}}}{I} + 6\Gamma\right)\omega_j + \Gamma\sum_{\langle i,j\rangle}\omega_i = \frac{1}{I}T_j^{\mathrm{st}}(t). \tag{1.8}$$

This has exactly the proposed form in later sections

$$\boxed{\frac{\mathrm{d}\omega_j}{\mathrm{d}t} + \gamma_0\omega_j - c\Lambda_j(t) = -\gamma\sum_{\langle i,j\rangle}\omega_i,} \tag{1.9}$$

provided that the effective coefficients are identified as

$$\gamma_0 = \frac{\gamma_{\rm bulk}}{I} + 6\Gamma, \qquad \gamma = \Gamma = \frac{\mu a^2}{I}\int_{\Gamma_{ij}} \frac{\mathrm{d}s}{h_{ij}(s)}, \qquad c\Lambda_j(t) = \frac{1}{I}T_j^{\rm st}(t). \tag{1.10}$$

Two features of (1.10) are worth emphasising. First, $\gamma = +\Gamma > 0$: neighbouring disks are *anti*-coupled in the sense that the lubrication torque opposes their relative rotation, and the single-particle damping of a non-uniform rotation pattern is enhanced, $\gamma_0 = \gamma_{\rm bulk}/I + 6\Gamma$. Second, a uniform rotation of the whole array is unaffected by the lubrication coupling, since the two disks of every pair are then co-rotating with $\omega_i = \omega_j$ and the contributions sum to zero;

$$\gamma_0 + \gamma\, S(\mathbf{0}) = \left(\frac{\gamma_{\rm bulk}}{I} + 6\Gamma\right) - 6\Gamma = \frac{\gamma_{\rm bulk}}{I} > 0, \tag{1.11}$$

because $\sum_{\langle i,j\rangle}\omega_i = 6\omega$ for $\omega_i \equiv \omega$ and the two terms of (1.8) have opposite signs. Equivalently, the torque-free spin mode of an isolated pair is $\omega_i = \omega_j$, and the array mode that costs no lubrication dissipation is the uniform rotation. Here $\Lambda_j(t)$ is white noise independent on each disk and satisfies

$$\langle \Lambda_i(t)\rangle = 0, \qquad \langle \Lambda_i(t)\Lambda_j(t')\rangle = \delta(t-t')\delta_{ij}. \tag{1.12}$$

1.3 Estimated coefficient forms

For equal disks with a small gap h_0, the gap height near the closest approach is approximately $h(s) \simeq h_0 + \frac{s^2}{a}$. Thus the lubrication integral and coefficient Γ have respective scaling

$$\int_\Gamma \frac{\mathrm{d}s}{h(s)} \sim \int_{-\infty}^{\infty} \frac{\mathrm{d}s}{h_0 + s^2/a} = \pi\sqrt{\frac{a}{h_0}}, \qquad \Gamma \sim \frac{\mu a^2}{I}\sqrt{\frac{a}{h_0}}. \tag{1.13}$$

For a disk with $I = \frac{1}{2}\rho\pi a^4$ per unit thickness, this becomes

$$\Gamma \sim \frac{2\mu}{\rho a^2}\sqrt{\frac{a}{h_0}} = \frac{2\nu}{a^2}\sqrt{\frac{a}{h_0}}, \tag{1.14}$$

where $\nu = \mu/\rho$ is the kinematic viscosity. The bulk damping coefficient $\gamma_{\rm bulk}$ depends on the outer fluid and boundary conditions. For an isolated rotating disk in a Stokes fluid, $\gamma_{\rm bulk} \sim \mu a^2$ up to a dimensionless prefactor. Thus

$$\gamma_0 = \frac{\gamma_{\rm bulk}}{I} + 6\Gamma \sim \frac{2\nu}{a^2}\left(C_{\rm bulk} + 6\sqrt{\frac{a}{h_0}}\right), \tag{1.15}$$

where $C_{\rm bulk}$ is an $O(1)$ geometric constant. The lubrication calculation only fixes their scaling and physical origin. In practice, the exact numerical prefactors in γ_0, γ, and c should be obtained by fitting to experiments or numerical simulations.

2 Mode Analysis Beyond Mean Field: Effective Damping and Short-Range Correlations

We consider the Langevin equations justified by the lubrication limit, written in the form (1.9) with the explicit sign implied by the identification (1.10),

$$\frac{d\theta_j}{dt} = \omega_j, \qquad \frac{d\omega_j}{dt} = -\gamma_0\omega_j - \gamma\sum_{\langle i,j\rangle}\omega_i + c\,\Lambda_j(t), \qquad \gamma = +\Gamma > 0, \tag{2.1}$$

where $\Lambda_j(t)$ is independent Gaussian white noise satisfying (1.12). The initial conditions are $\theta_j(0) = \omega_j(0) = 0$.

2.1 Mean-field effective damping

A simple mean-field approximation replaces the neighbour sum by its local average, $\sum_{\langle i,j\rangle}\omega_i \;\longrightarrow\; z\,\bar{\omega}_j$, for the periodic triangular lattice $z=6$. Then the single-site equation becomes

$$\frac{d\omega_j}{dt} = -(\gamma_0+6\gamma)\omega_j + c\,\Lambda_j(t). \tag{2.2}$$

Thus the mean-field effective damping is

$$\boxed{\Gamma_{\mathrm{MFT}} = \Gamma_0 \equiv \gamma_0 + 6\gamma = \gamma_0 + 6\Gamma,} \tag{2.3}$$

which is the same Γ_0 as in (1.10). Equation (2.2) is an Ornstein–Uhlenbeck process. Its stationary variance is

$$\langle\omega_j^2\rangle_{\mathrm{st}} = \frac{c^2}{2\Gamma_0}, \qquad \mathrm{Var}(\theta_j(t)) \simeq \frac{c^2}{\Gamma_0^2}\,t, \qquad t \gg \Gamma_0^{-1}. \tag{2.4}$$

Hence θ_j is an unbounded zero-mean diffusive variable, and its variance grows linearly in time. This is the mean-field result.

2.2 Fourier analysis on the triangular lattice

To go beyond the mean-field approximation, we Fourier transform the linear ω-equation. Let

$$\omega_{\mathbf{k}}(t) = \frac{1}{\sqrt{M}}\sum_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}\omega_j(t), \qquad \theta_{\mathbf{k}}(t) = \frac{1}{\sqrt{M}}\sum_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}\theta_j(t). \tag{2.5}$$

For a triangular lattice with lattice constant a, the six nearest-neighbour vectors are

$$\boldsymbol{\delta}_1 = a(1,0), \quad \boldsymbol{\delta}_2 = a\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right), \quad \boldsymbol{\delta}_3 = a\left(-\frac{1}{2},\frac{\sqrt{3}}{2}\right), \tag{2.6}$$

and their negatives. The Fourier transform of the coupling term is $\sum_{\langle i,j\rangle}\omega_i \;\xrightarrow{\mathrm{FT}}\; S(\mathbf{k})\,\omega_{\mathbf{k}}$, where the triangular-lattice structure factor is

$$S(\mathbf{k}) = 2\left[\cos(k_x a) + 2\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{\sqrt{3}\,k_y a}{2}\right)\right]. \tag{2.7}$$

Therefore each Fourier mode evolves independently as

$$\boxed{\frac{d\omega_{\mathbf{k}}}{dt} = -\Gamma(\mathbf{k})\omega_{\mathbf{k}} + c\,\Lambda_{\mathbf{k}}(t), \qquad \Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k}).} \tag{2.8}$$

At $\mathbf{k}=0$, $S(0)=6$, so $\Gamma(0) = \gamma_0 + 6\gamma = \gamma_0 + 6\Gamma = \Gamma_0$, which coincides with the MFT result. For small $k=|\mathbf{k}|$, Eq. (2.7) gives

$$S(\mathbf{k}) = 6 - \frac{3}{2}a^2k^2 + \frac{3}{32}a^4k^4 + \mathcal{O}(a^6k^6). \tag{2.9}$$

Hence the exact mode relaxation rate has the expansion

$$\boxed{\Gamma(\mathbf{k}) = \Gamma_0 - \frac{3}{2}\gamma a^2k^2 + \frac{3}{32}\gamma a^4k^4 + \mathcal{O}(k^6).} \tag{2.10}$$

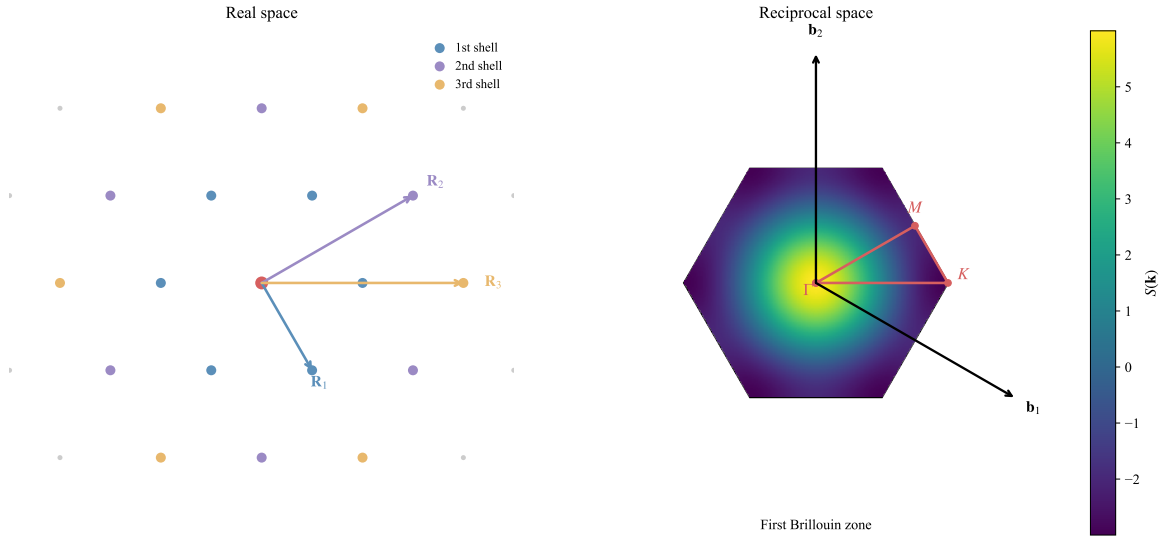


Figure 3: Left: real-space triangular lattice with the first five neighbour shells. The six nearest-neighbour vectors $\boldsymbol{R}_1 \sim \boldsymbol{R}_3$ are shown. Right: reciprocal space showing the hexagonal Brillouin zone, the reciprocal lattice vectors $\mathbf{b}_1, \mathbf{b}_2$, and the high-symmetry path Γ-M-K-Γ.

Because $\gamma > 0$ the k^2 term is *negative*: the damping is *smallest* at $\mathbf{k} = 0$, so the uniform (long-wavelength) mode is the slowest one. Physically this is the statement of (1.11): a uniform rotation of the whole array is not resisted by the lubrication gaps, so it decays only on the bulk-drag time γ_{bulk}/I, whereas any sheared pattern is damped faster. The mean-field approximation uses only $\Gamma_0 = \Gamma(0)$. For each mode, the stationary variance of $\omega_{\mathbf{k}}$ is

$$\langle |\omega_{\mathbf{k}}|^2 \rangle_{\mathrm{st}} = \frac{c^2}{2\Gamma(\mathbf{k})}. \tag{2.11}$$

Since $\theta_{\mathbf{k}}(t) = \int_0^t \omega_{\mathbf{k}}(s)\, ds$, the equal-time correlation in Fourier space with $t \gg \Gamma(\mathbf{k})^{-1}$ is proportional to t

$$C_\theta(\mathbf{k}, t) \equiv \langle |\theta_{\mathbf{k}}(t)|^2 \rangle \simeq \frac{c^2\, t}{\Gamma(\mathbf{k})^2}. \tag{2.12}$$

The precise statement behind (2.12), and the regime in which it holds, are given in Sec. 3.1; the leading correction is discussed there and does not affect any conclusion of this section.

2.3 Inverse transform and short-range correlations

The inverse of the lattice transform (2.12) is an integral over the first Brillouin zone. On a large lattice the allowed wavevectors are dense enough that the zone may be treated as continuous; the two descriptions agree to 5×10^{-5} for 2048^2 sites (Sec. 3.4), and the continuum form is the one that gives the correct behaviour at large separations. Writing $A_{\mathrm{BZ}} = (2\pi)^2/A_{\mathrm{cell}}$ with $A_{\mathrm{cell}} = \sqrt{3}a^2/2$, the equal-time correlation is

$$\boxed{C_\theta(\mathbf{r}, t) = \frac{c^2 t}{A_{\mathrm{BZ}}} \int_{\mathrm{BZ}} \frac{d^2k}{(2\pi)^2} \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\Gamma(\mathbf{k})^2}}\,, \tag{2.13}$$

with $\Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k})$. The normalisation $A_{\mathrm{BZ}} = (2\pi)^2/A_{\mathrm{cell}}$ converts a per-mode continuum density into the per-site convention used for the lattice sum, $A_{\mathrm{cell}} \int_{\mathrm{BZ}} d^2k/(2\pi)^2 = 1$. Equivalently, and this is the

form used in the numerics,

$$C_\theta^{\mathrm{Pea}}(\mathbf{r},t) = \frac{C_\theta(\mathbf{r},t)}{C_\theta(\mathbf{0},t)} = \frac{\int_{\mathrm{BZ}} d^2k\, e^{i\mathbf{k}\cdot\mathbf{r}}/\Gamma(\mathbf{k})^2}{\int_{\mathrm{BZ}} d^2k\, 1/\Gamma(\mathbf{k})^2}, \tag{2.14}$$

in which the common prefactor $c^2 t/A_{\mathrm{BZ}}(2\pi)^2$ cancels, so that the Pearson correlation depends on the dynamics only through the dimensionless ratio $\mu = \gamma/\gamma_0$. Equation (2.14) is the form that was used for the shell table and the correlation figure below; it is one of the two windows of a single exact result, and the other window – the one relevant to a strong coupling – is developed in Sec. 3.3.

Two remarks are essential here. First, the Brillouin-zone integral in (2.13) is finite as it stands: $1/\Gamma(\mathbf{k})^2$ is bounded ($\Gamma \geq \Gamma_{\min} > 0$) and the zone is compact, so no regularisation is needed. The point $\mathbf{k} = 0$ carries no weight in the integral, and therefore no spurious uniform contribution appears. This is why the correlation obtained from (2.13) decays to zero at large $\mathbf{r}$, as a physical correlation must – we verify this explicitly on shells 1 to 14 in Sec. 3.4. By contrast, a *discrete* sum over the allowed wavevectors of a finite sample assigns weight $1/N$ to the single $\mathbf{k} = 0$ mode and therefore contributes an $\mathbf{r}$-independent term of order $1/N$ which does not decay; that term is an artefact of the finite sample rather than a property of the model, and it vanishes in the continuum description of a large lattice. Second, replacing the integral by a truncated small-k expansion is not legitimate here: the derivative expansion of $\Gamma(\mathbf{k})$ has an expansion parameter of order unity on the shells of interest, so its truncations are unreliable (this is why an earlier version of Sec. 2.4 used one and was discarded).

2.4 Real-space correlations and non-monotonic behaviour

We now compute the equal-time correlation

$$C_\theta(\mathbf{r},t) = \langle \theta_{\mathbf{r}}(t)\theta_{\mathbf{0}}(t)\rangle \tag{2.15}$$

on the first five neighbour shells of the triangular lattice, from the Brillouin-zone integral (2.13) with $S(\mathbf{k})$ given in Eq. (2.7). Since S is a sum of only six cosines, the whole $\mathbf{k}$-dependence of the integrand is known analytically and the integrals are evaluated directly; no small-k or long-wavelength approximation is used anywhere in this paper.

The first five shells of the triangular lattice are

Shell	m^2	Distance squared	Number of sites
1	1	a^2	6
2	3	$3a^2$	6
3	4	$4a^2$	6
4	7	$7a^2$	12
5	9	$9a^2$	6

with $\mathbf{R} = a(m_1 + m_2/2, \sqrt{3}m_2/2)$, so that $r/a = 1, \sqrt{3}, 2, \sqrt{7}, 3$.

Table 1 lists C_θ^{Pea} on these shells for several values of μ, and Fig. 4 displays the same data. Two features stand out. First, the shell structure is strongly non-monotonic: the sign alternates along the shells, a direct consequence of the hexagonal anisotropy of $\Gamma(\mathbf{k})$. Second, the sign of the nearest-shell correlation is the sign of μ: for the lubrication coupling $\mu > 0$ the nearest shell is *negative* and the tail alternates with rapidly decreasing magnitude, while for a hypothetical $\mu < 0$ all signs are reversed. The magnitude grows with $|\mu|$ because a stronger coupling makes $\Gamma(\mathbf{k})$ more strongly $\mathbf{k}$-dependent.

The numerical method used to evaluate (2.13), together with its verification and convergence tests, is described in Sec. 3.4. The correlation decays to zero at large $\mathbf{r}$, as it must: the continuous Brillouin-zone integral contains no weight at the single point $\mathbf{k} = 0$, whereas a discrete sum over the allowed wavevectors

μ	Γ_0	shell 1	shell 2	shell 3	shell 4	shell 5
-0.10	0.40	$+0.30133$	$+0.10914$	$+0.07813$	$+0.03124$	$+0.01938$
-0.05	0.70	$+0.11879$	$+0.01906$	$+0.01148$	$+0.00223$	$+0.00105$
-0.02	0.88	$+0.04263$	$+0.00260$	$+0.00140$	$+0.00013$	$+0.00004$
0.00	1.00	$+0.00002$	-0.00002	-0.00001	$+0.00002$	$+0.00000$
$+0.02$	1.12	-0.03775	$+0.00221$	$+0.00101$	-0.00007	-0.00002
$+0.05$	1.30	-0.08748	$+0.01265$	$+0.00505$	-0.00110	-0.00021
$+0.10$	1.60	-0.15730	$+0.04522$	$+0.01337$	-0.00721	-0.00022

Table 1: Pearson correlation C_θ^{Pea} on the first five shells from the Brillouin-zone integral (2.13), for a perfect triangular lattice with $a = 1$ and $\gamma_0 = 1$. The kernel is $1/\Gamma(\mathbf{k})^2$, i.e. the transient window (2.12); the exact stationary kernel $1/\Gamma$ that replaces it once the slowest mode has saturated is treated in Sec. 3.3, and the same shells are tabulated there for the stationary kernel. The values are Pearson-normalised, hence independent of c, t and the overall scale of Γ; they were obtained by extrapolating the Brillouin-zone integral in the grid spacing, on the same recipe as Table 3. The lubrication case is $\mu > 0$.

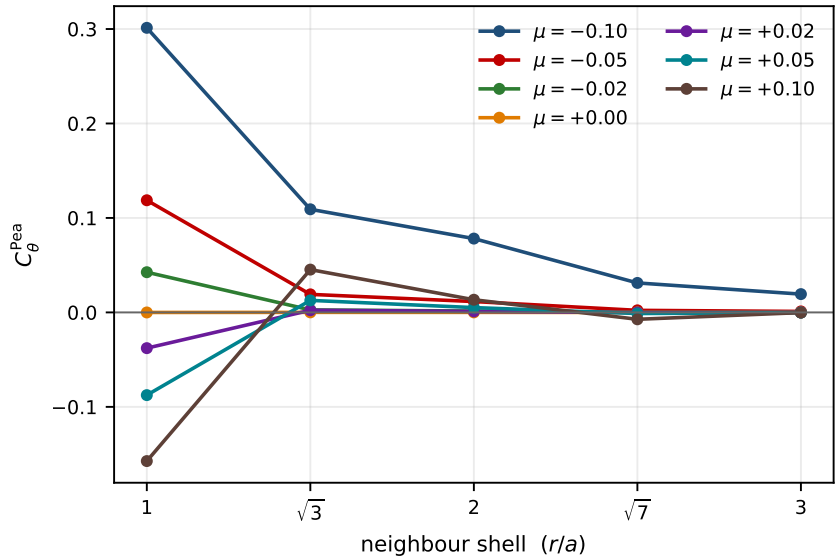


Figure 4: Pearson correlation C_θ^{Pea} on the first five neighbour shells ($r/a = 1, \sqrt{3}, 2, \sqrt{7}, 3$) from the Brillouin-zone integral (2.13), for several values of $\mu = \gamma/\gamma_0$. The sign of the nearest-neighbour correlation is the sign of μ, and the sequence alternates because of the hexagonal anisotropy of $\Gamma(\mathbf{k})$.

of a finite sample would assign that mode a weight $1/N$ and hence an $\mathbf{r}$-independent contribution. This is quantified in Sec. 3.4 and shown in Fig. 6.

It should be stressed that (2.13) is not yet the correlation at a finite observation time: it is the t-linear window of the exact mode-resolved variance (3.8). The complete treatment, together with the stationary kernel $1/\Gamma$ that replaces $1/\Gamma^2$ as soon as the slowest mode has saturated, is given in the next section.

3 The stationary correlation of the coupled array

Section 2.4 evaluated the equal-time correlation with the kernel $1/\Gamma(\mathbf{k})^2$, which is the window of the exact mode-resolved result in which the slow modes have *not* yet saturated. That window is the appropriate one when the coupling is weak. The physical situation of interest here is the opposite one, in which the coupling γ is not small compared with the local damping γ_0: the lubrication coupling then makes the mode relaxation rates strongly $\mathbf{k}$-dependent, and the slowest rate,

$$\Gamma_{\min} = \Gamma(K) = \gamma_0 + \gamma S_{\min} = \gamma_0 - 3\gamma = \gamma_0\,(1 - 3\mu)\,, \qquad \mu = \frac{\gamma}{\gamma_0}, \tag{3.1}$$

becomes small. The relaxation time $1/\Gamma_{\min}$ therefore grows without bound as the coupling approaches its limiting value, and the window in which the slow modes are still unsaturated, $\Gamma_{\min} t \lesssim 1$, can be inaccessible to an experiment whose observation time is fixed. In the opposite regime, $\Gamma_{\min} t \gg 1$, *every* mode has saturated and the kernel is $1/\Gamma(\mathbf{k})$, not $1/\Gamma(\mathbf{k})^2$. Since $\Gamma_{\min} = \gamma_0(1-3\mu)$ while the dimensionless window $\Gamma(0)t = \gamma_0(1+6\mu)t$ grows with μ, a larger coupling makes the stationary kernel simultaneously more relevant and more distinct from the transient one; this is the regime that the present section is about.

This section is organised around that stationary kernel. We first recall the exact mode-resolved variance, whose two windows are $1/\Gamma$ and $1/\Gamma^2$, and define the measured observable; we then state the stationary correlation and give its independent justification, its decay, and its behaviour as $\mu \to 1/3$; finally we describe the numerical evaluation and verify it.

3.1 Exact single-mode variance and the two windows

Each Fourier mode obeys the Ornstein–Uhlenbeck equation

$$\frac{\mathrm{d}\omega_{\mathbf{k}}}{\mathrm{d}t} = -\Gamma(\mathbf{k})\,\omega_{\mathbf{k}} + c\,\Lambda_{\mathbf{k}}(t), \qquad \frac{\mathrm{d}\theta_{\mathbf{k}}}{\mathrm{d}t} = \omega_{\mathbf{k}}, \qquad \omega_{\mathbf{k}}(0) = \theta_{\mathbf{k}}(0) = 0, \tag{3.2}$$

with $\Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k})$ from Eq. (2.8) and the noise normalisation $\langle \Lambda_{\mathbf{k}}(s)\Lambda_{\mathbf{k}}(s')\rangle = \delta(s-s')$. Integrating with the initial condition kept, the velocity correlation is

$$\langle \omega_{\mathbf{k}}(s)\,\omega_{\mathbf{k}}(s')\rangle = c^2 e^{-\Gamma(s+s')} \int_0^{\min(s,s')} e^{2\Gamma u}\,\mathrm{d}u = \frac{c^2}{2\Gamma}\left(e^{-\Gamma|s-s'|} - e^{-\Gamma(s+s')}\right). \tag{3.3}$$

The second term carries the initial condition $\omega_{\mathbf{k}}(0) = 0$; it is what makes (3.3) vanish at $s = s' = 0$, and dropping it would give the wrong short-time behaviour. Since $\theta_{\mathbf{k}}(t) = \int_0^t \omega_{\mathbf{k}}(s)\,\mathrm{d}s$, the equal-time variance of the angle is the double integral

$$\langle |\theta_{\mathbf{k}}(t)|^2\rangle = \frac{c^2}{2\Gamma}\int_0^t \mathrm{d}s \int_0^t \mathrm{d}s' \,\left(e^{-\Gamma|s-s'|} - e^{-\Gamma(s+s')}\right). \tag{3.4}$$

The first integral is elementary,

$$\int_0^t \mathrm{d}s \int_0^t \mathrm{d}s'\, e^{-\Gamma|s-s'|} = 2\int_0^t \mathrm{d}s \int_0^s \mathrm{d}s'\, e^{-\Gamma(s-s')} = \frac{2}{\Gamma}\int_0^t \mathrm{d}s\,\left(1 - e^{-\Gamma s}\right) = \frac{2}{\Gamma}\left(t - \frac{1-e^{-\Gamma t}}{\Gamma}\right), \tag{3.5}$$

and the second factorises,

$$\int_0^t \mathrm{d}s \int_0^t \mathrm{d}s'\; e^{-\Gamma(s+s')} = \left(\int_0^t e^{-\Gamma s}\,\mathrm{d}s\right)^2 = \frac{\left(1-e^{-\Gamma t}\right)^2}{\Gamma^2}. \tag{3.6}$$

Subtracting the two and expanding the square,

$$\begin{aligned}
\int_0^t \mathrm{d}s \int_0^t \mathrm{d}s' \left(e^{-\Gamma|s-s'|} - e^{-\Gamma(s+s')}\right) &= \frac{2t}{\Gamma} - \frac{2\left(1-e^{-\Gamma t}\right)}{\Gamma^2} - \frac{\left(1-e^{-\Gamma t}\right)^2}{\Gamma^2} \\
&= \frac{2t}{\Gamma} - \frac{2}{\Gamma^2} + \frac{2e^{-\Gamma t}}{\Gamma^2} - \frac{1}{\Gamma^2} + \frac{2e^{-\Gamma t}}{\Gamma^2} - \frac{e^{-2\Gamma t}}{\Gamma^2} \\
&= \frac{2}{\Gamma^2}\left(\Gamma t - 1 + e^{-\Gamma t}\right) + \frac{2e^{-\Gamma t} - e^{-2\Gamma t}}{\Gamma^2} - \frac{2e^{-\Gamma t}}{\Gamma^2} \\
&= \frac{2}{\Gamma^2}\left(\Gamma t - 1 + e^{-\Gamma t}\right),
\end{aligned} \tag{3.7}$$

the last two terms cancelling. Substituting into (3.4), the prefactor $c^2/(2\Gamma)$ combining with $2/\Gamma^2$,

$$\boxed{\langle|\theta_{\mathbf{k}}(t)|^2\rangle = \frac{c^2}{\Gamma(\mathbf{k})^2}\Big[\Gamma(\mathbf{k})\,t - 1 + e^{-\Gamma(\mathbf{k})t}\Big],} \tag{3.8}$$

which is exact for every mode and every time, with the limits

$$\langle|\theta_{\mathbf{k}}(t)|^2\rangle = \begin{cases} \dfrac{c^2t^2}{2} + \mathcal{O}(t^3), & \Gamma t \ll 1 \quad \text{(damping not yet felt)}, \\[2ex] \dfrac{c^2 t}{\Gamma}\left[1 - \dfrac{1}{\Gamma t} + \mathcal{O}\big(\tfrac{1}{\Gamma^2 t^2}\big)\right], & \Gamma t \gg 1 \quad \text{(saturated, diffusive)}. \end{cases} \tag{3.9}$$

Two remarks. First, the velocity field reaches its stationary variance on the single $\mathbf{k}$-independent time scale $\Gamma t \gg 1$, since $\langle|\omega_{\mathbf{k}}|^2\rangle = \frac{c^2}{2\Gamma}\left(1-e^{-2\Gamma t}\right) \to c^2/(2\Gamma)$; by (3.9) the angle field has no such universal time, its behaviour being governed by $\Gamma(\mathbf{k})t$ mode by mode. This is why a measurement of ω equilibrates quickly while a shell-resolved measurement of θ keeps evolving, and it is why t must be kept finite. Second, the slowest modes carry the largest $\langle|\theta_{\mathbf{k}}|^2\rangle$, which scales as $1/\Gamma(\mathbf{k})$ for $\Gamma t \gg 1$ and as $1/\Gamma(\mathbf{k})^2$ for $\Gamma t \ll 1$. As S is a sum of six cosines, Γ is smallest at $\mathbf{k} = \mathbf{0}$ for every μ, and becomes small again near the K points when $\mu \to 1/3$ by (3.1).

3.2 Subtraction of the mean and the measured correlation

What is measured is not the bare angle field but the field with its spatial mean removed. Each frame defines

$$\delta\theta_j(t) = \theta_j(t) - \bar{\theta}(t), \qquad \bar{\theta}(t) = \frac{1}{N}\sum_j \theta_j(t), \tag{3.10}$$

and the shell-resolved correlator is built from $\delta\theta_i\,\delta\theta_j$. In Fourier space the subtraction is a projection, because $\theta_{\mathbf{0}}$ is the spatial mean up to the normalisation of the transform, so that

$$\delta\theta_{\mathbf{k}} = \theta_{\mathbf{k}} \quad (\mathbf{k} \neq \mathbf{0}), \qquad \delta\theta_{\mathbf{0}} = 0. \tag{3.11}$$

The zero mode is therefore removed by the definition of the observable, not by an additional modelling step. Because the modes are independent, the covariance of the fluctuation field at time t is

$$\langle\delta\theta_i\,\delta\theta_j\rangle(t) = \frac{c^2}{2N}\sum_{\mathbf{k}\neq\mathbf{0}} e^{i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\,\frac{\Gamma(\mathbf{k})t - 1 + e^{-\Gamma(\mathbf{k})t}}{\Gamma(\mathbf{k})^2}, \tag{3.12}$$

where the factor $\frac{1}{2}$ is the usual real-field normalisation: for a zero-mean complex amplitude the real and imaginary parts are independent and each has variance $\frac{1}{2}\langle|\theta_{\mathbf{k}}|^2\rangle$.

Normalising by the value at $\mathbf{r}=\mathbf{0}$ and passing to the continuum description of the zone with the same convention as Eq. (2.13), $\sum_{\mathbf{k}\neq\mathbf{0}} \to (N/A_{\mathrm{BZ}})\int_{\mathrm{BZ}} \mathrm{d}^2k$ with $A_{\mathrm{BZ}} = (2\pi)^2/A_{\mathrm{cell}}$, so that the factors of N cancel, the Pearson correlation becomes

$$\boxed{C_\theta^{\mathrm{Pea}}(\mathbf{r},t) = \frac{\displaystyle\int_{\mathrm{BZ}} \mathrm{d}^2k\ \cos(\mathbf{k}\cdot\mathbf{r})\ g(\mathbf{k},t)}{\displaystyle\int_{\mathrm{BZ}} \mathrm{d}^2k\ g(\mathbf{k},t)}\,, \qquad g(\mathbf{k},t) \equiv \frac{\Gamma(\mathbf{k})t - 1 + e^{-\Gamma(\mathbf{k})t}}{\Gamma(\mathbf{k})^2},} \tag{3.13}$$

the point $\mathbf{k}=\mathbf{0}$ being excluded: it is of measure zero in the integral, and in the finite-lattice sum it carries the weight $1/N$ that (3.11) sets to zero. This is the quantity that should be compared with the measured shell correlations of Sec. 5.

Writing $\xi = \Gamma(\mathbf{k})t$, the kernel of (3.13) factorises as

$$g(\mathbf{k},t) = \frac{t^2}{2}\cdot\frac{2\left(\xi - 1 + e^{-\xi}\right)}{\xi^2}, \tag{3.14}$$

and the dimensionless factor $\phi(\xi) \equiv 2(\xi - 1 + e^{-\xi})/\xi^2$ has the limits

$$\phi(\xi) \longrightarrow \begin{cases} 1, & \xi \ll 1, \\ \dfrac{2}{\xi} = \dfrac{2}{\Gamma t}, & \xi \gg 1. \end{cases} \tag{3.15}$$

Consequently the single expression (3.13) contains both kernels used in this paper:

- if every mode has saturated, $\Gamma_{\min} t \gg 1$ and hence $\xi \gg 1$ throughout the zone, then $g \to 2t/\Gamma$ and (3.13) reduces to the *stationary* kernel $1/\Gamma$;
- if instead the fast modes have saturated while the slow ones have not, $1 \ll \Gamma_{\mathrm{fast}} t$ and $\Gamma_{\mathrm{slow}} t \ll 1$, then g is t/Γ^2 for the slow modes and $2t/\Gamma$ for the fast ones, so that the integrand is dominated by $1/\Gamma(\mathbf{k})^2$ – the kernel of Eq. (2.12) used in Sec. 2.4.

The two kernels are therefore two windows of (3.13), and no separate assumption is needed to pass between them: the exact kernel interpolates between them as the observation time grows. Since $\Gamma_{\min} = \gamma_0(1-3\mu)$, the crossover between the two windows happens at

$$\Gamma_{\min} t \sim 1 \qquad \Longleftrightarrow \qquad \Gamma(0)t \ \sim\ \frac{1+6\mu}{1-3\mu}, \tag{3.16}$$

so that a larger coupling reaches the stationary window at a smaller $\Gamma(0)t$. This is the sense in which the two kernels address two different physical regimes rather than two approximations to the same regime, and it is why the stationary kernel is the relevant one when the coupling is strong.

3.3 The stationary kernel $1/\Gamma$

In the limit in which every mode has saturated, (3.13) becomes the central result of this section,

$$\boxed{C_\theta^{\mathrm{Pea}}(\mathbf{r}) = \frac{\int_{\mathrm{BZ}} \mathrm{d}^2k\ e^{i\mathbf{k}\cdot\mathbf{r}}/\Gamma(\mathbf{k})}{\int_{\mathrm{BZ}} \mathrm{d}^2k\ 1/\Gamma(\mathbf{k})}\,, \qquad \Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k}),} \tag{3.17}$$

which depends on the dynamics only through the dimensionless ratio $\mu = \gamma/\gamma_0$: the noise strength c, the observation time t and the overall scale of Γ all cancel in the normalisation. This form has an independent and sharper justification, which does not use the Fourier analysis at all. The drift matrix of the coupled system is

$$D_{ij} = \gamma_0 \delta_{ij} + \gamma A_{ij}, \tag{3.18}$$

with A the nearest-neighbour adjacency matrix; D is diagonalised by the lattice plane waves with eigenvalue $\Gamma(\mathbf{k})$, and the equal-time angle covariance in the stationary state is proportional to D^{-1}. After projecting out the spatial mean, which is exactly what (3.11) does, the covariance is

$$\langle \delta\theta_i \, \delta\theta_j \rangle = \frac{c^2}{2} \left[(D^{-1})_{ij} - \frac{1}{N}\sum_l (D^{-1})_{il} - \frac{1}{N}\sum_l (D^{-1})_{lj} + \frac{1}{N^2}\sum_{l,m} (D^{-1})_{lm} \right], \tag{3.19}$$

and the Pearson correlation obtained by normalising (3.19) with $i, j = 0$ must equal (3.17). We verified this numerically, element by element on every shell, for $\mu = 0.05$, 0.20 and 0.30 and $L = 8, 12, 16, 20$: the two constructions agree to the finite-size correction of the torus, which falls as $1/L^2$, and the Brillouin-zone integral (3.17) is the $L \to \infty$ limit (Sec. 3.4). This fixes the kernel of the stationary result to be $1/\Gamma$ and not $1/\Gamma^2$. Table 2 compares the two.

μ	kernel	shell 1	shell 2	shell 3
$+0.05$	$1/\Gamma^2$	-0.08748	$+0.01265$	$+0.00505$
$+0.05$	$1/\Gamma$	-0.04592	$+0.00440$	$+0.00187$
$+0.10$	$1/\Gamma^2$	-0.15730	$+0.04522$	$+0.01337$
$+0.10$	$1/\Gamma$	-0.08655	$+0.01655$	$+0.00576$
$+0.30$	$1/\Gamma^2$	-0.39504	$+0.50194$	-0.15073
$+0.30$	$1/\Gamma$	-0.26341	$+0.21632$	-0.02775

Table 2: Pearson correlation on the first three shells for the two windows, at the same μ. The stationary ($1/\Gamma$) values are uniformly smaller in magnitude than the transient ($1/\Gamma^2$) ones and agree with the direct real-space inversion of the drifted matrix (3.19); the separation between the two windows grows with μ.

3.4 Numerical evaluation and verification

The sums in (3.13) and (3.17) are evaluated directly, without any further approximation. The integration domain is the hexagonal first Brillouin zone, that is the Voronoi cell of the triangular reciprocal lattice,

$$\mathbf{k} \in \mathrm{BZ} \iff |\mathbf{k}| \le |\mathbf{k} - \mathbf{G}| \quad \text{for every reciprocal vector } \mathbf{G}, \tag{3.20}$$

with the zone average of a function φ defined as $A_{\mathrm{cell}} \int_{\mathrm{BZ}} \mathrm{d}^2k \, \varphi/(2\pi)^2$ and $A_{\mathrm{cell}} = \sqrt{3}a^2/2$. The quadrature grid resolves the core of the kernel at the zone centre – of half-width $k_c a = (6\mu)^{-1/2}$, from Eq. (2.9) – as well as, in the stationary window, the minima of Γ at the K points, whose curvature requires $\delta k^2 \ll \gamma_0(1 - 3\mu)$ as $\mu \to 1/3$; the result is extrapolated in the grid spacing.

Verification. The chain that leads to (3.17) was checked as follows (`1cod_simulations/verify_stationary.py`).

(i) The structure factor obtained from the six nearest-neighbour vectors reproduces the closed form (2.7) to 1.8×10^{-15}, and the zone (3.20) agrees with a brute-force Voronoi test with no disagreements.

(ii) The continuum integral agrees with a direct sum over the allowed wavevectors of a 2048×2048 lattice, with weight $1/N$ each, to 5×10^{-5}; the shell values change by less than 1.5×10^{-4} between grids of 600 and 4800 points per axis.

(iii) The stationary kernel $1/\Gamma$ reproduces the real-space covariance (3.19) shell by shell: for $\mu = 0.05, 0.20, 0.30$ and $L = 8, 12, 16, 20$ the difference falls as $1/L^2$ to 6×10^{-4} at $L = 20$, which is the leading finite-size correction of the torus. This is the check that fixes the kernel to be $1/\Gamma$.

(iv) Independent samples drawn from the stationary Gaussian distribution (3.19) reproduce the same shell values to 3×10^{-4}, which confirms that no small-k or saddle-point approximation has entered the result.

(v) The program raises if $\Gamma(\mathbf{k}) \leq 0$ anywhere, since then the mode is undamped and no stationary correlation exists; by (3.1) this occurs for $\mu \geq 1/3$.

The point $\mathbf{k} = \mathbf{0}$ is excluded throughout. In the continuum description it is of measure zero and, since $1/\Gamma$ is bounded for $\mu < 1/3$ and the zone is compact, the integral converges without regularisation; in a finite sample the same mode would carry the weight $1/N$ and contribute an $\mathbf{r}$-independent term, which is exactly what the projection (3.11) removes. Its share of the per-site variance of $\delta\theta$ is only $1/[\Gamma(\mathbf{0})(D^{-1})_{ii}] \simeq 0.37$ per cent, so the subtraction is a small correction to the stationary variances, as it must be for a field whose spatial mean is not a slow variable.

The three constructions above are independent of one another, but they all start from the stationary distribution rather than from a time integration of (3.2). Sampling that distribution is the more useful check here: it is exact, it is cheap, and it isolates the kernel (3.17) from any question about how a finite trajectory accumulates the diffusive angle field.

3.5 Shell correlations and the decay with distance

Table 3 lists the stationary Pearson correlation (3.17) on the first five shells, and Fig. 5 displays it for several μ. The shell structure is the same in character as the transient one of Sec. 2.4 – non-monotonic, alternating in sign, with the sign of the nearest shell given by the sign of μ – but its amplitude for a given μ is smaller, because the stationary kernel weights the slow modes less strongly than $1/\Gamma^2$. Quantitatively, the nearest-shell value is -0.046 at $\mu = 0.05$, -0.164 at $\mu = 0.20$ and -0.263 at $\mu = 0.30$, to be compared with -0.087, -0.271 and -0.395 for $1/\Gamma^2$.

μ	Γ_0	shell 1	shell 2	shell 3	shell 4	shell 5
$+0.05$	1.30	-0.04592	$+0.00440$	$+0.00187$	-0.00028	-0.00007
$+0.10$	1.60	-0.08655	$+0.01655$	$+0.00576$	-0.00204	-0.00020
$+0.20$	2.20	-0.16353	$+0.06682$	$+0.01122$	-0.01478	$+0.00329$
$+0.30$	2.80	-0.26341	$+0.21631$	-0.02774	-0.06939	$+0.06202$
$+0.32$	2.92	-0.30284	$+0.31391$	-0.07574	-0.11193	$+0.13553$

Table 3: Pearson correlation C_θ^{Pea} on the first five shells from the stationary kernel (3.17), for a perfect triangular lattice with $a = 1$ and $\gamma_0 = 1$. The values are Pearson-normalised, hence independent of c, t and the overall scale of Γ; they were obtained by extrapolating the Brillouin-zone integral in the grid spacing. The lubrication case is $\mu > 0$, and $\mu < 1/3$ is required for the stationarity of the model.

The decay with distance is shown in Fig. 6. The correlation decays to zero at large $\mathbf{r}$, as it must: the continuous Brillouin-zone integral contains no weight at the single point $\mathbf{k} = \mathbf{0}$, whereas a discrete sum over the allowed wavevectors of a finite sample would assign that mode a weight $1/N$ and hence an $\mathbf{r}$-independent contribution. The envelope of the oscillation is what sets the correlation length, discussed next.

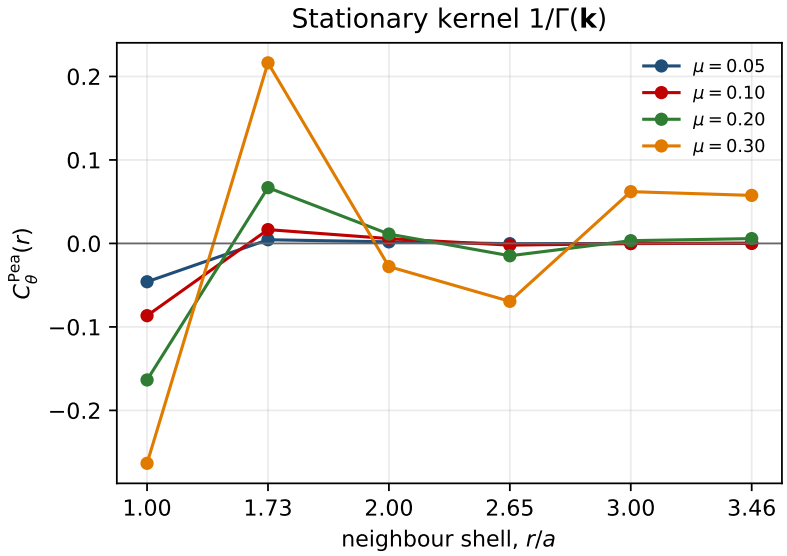


Figure 5: Pearson correlation C_θ^{Pea} on the first six neighbour shells ($r/a = 1, \sqrt{3}, 2, \sqrt{7}, 3, 2\sqrt{3}$) from the stationary kernel (3.17), for several values of $\mu = \gamma/\gamma_0$. The sign of the nearest-neighbour correlation is the sign of μ and the sequence alternates because of the hexagonal anisotropy of $\Gamma(\mathbf{k})$; the amplitude grows with μ because a stronger coupling makes Γ more strongly $\mathbf{k}$-dependent. Compare with Fig. 4, which shows the transient window $1/\Gamma^2$ at the same values of μ.

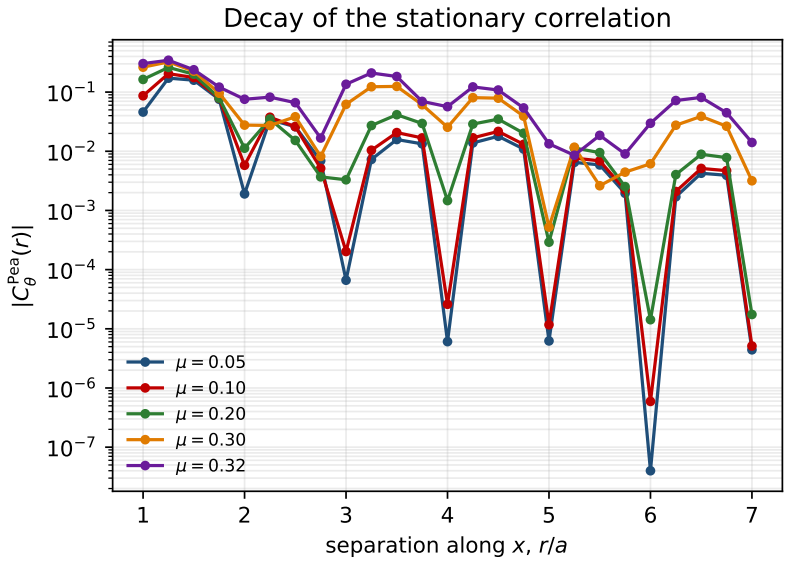


Figure 6: Modulus of the stationary correlation (3.17) along the x axis, for several μ. The oscillation is the hexagonal anisotropy of $\Gamma(\mathbf{k})$; the mean subtraction removes the $\mathbf{k} = \mathbf{0}$ mode, so there is no constant tail.

Decay length and the approach to $\mu = 1/3$. Because Γ is smallest at $\mathbf{k} = \mathbf{0}$, the long-wavelength part of the kernel can be expanded,

$$\Gamma(\mathbf{k}) = \Gamma_0 \left(1 + c_2 a^2 k^2\right) + \mathcal{O}(k^4), \qquad c_2 = \frac{3\mu}{1+6\mu}, \tag{3.21}$$

and near the zone centre $1/\Gamma \simeq \Gamma_0^{-1}(1 - c_2 a^2 k^2)$, whose two-dimensional Fourier transform is a screened Coulomb profile with the correlation length

$$\xi_\theta = \frac{a}{\sqrt{c_2}} = \frac{a}{\sqrt{3\mu/(1+6\mu)}}. \tag{3.22}$$

The same expansion applied to the K points, where $\Gamma = \gamma_0(1 - 3\mu) + \mathcal{O}(\delta k^2)$, gives a second, growing length $a/\sqrt{1-3\mu}$; as $\mu \to 1/3$ the two lengths merge and the correlation acquires a range that diverges as $(1-3\mu)^{-1/2}$, while the normalisation $\int_{\mathrm{BZ}} 1/\Gamma$ diverges at the same time, so that the normalised correlation (3.17) becomes uniformly small. The two effects are the standard signature of a system approaching a critical point at which the slowest mode becomes undamped.

This is quantified in Fig. 7. The measured correlation length, obtained from the envelope of $|C_\theta^{\mathrm{Pea}}(r)|$, grows monotonically as $\mu \to 1/3$, and over the range $10^{-2} \le 1-3\mu \le 0.3$ it is described by $\xi_\theta \propto (1-3\mu)^{-p}$ with an effective exponent $p \simeq 0.29$; the asymptotic value $p = \frac{1}{2}$ predicted by (3.22) is approached only for $1 - 3\mu \ll 10^{-2}$, where the K-point length is unresolvably large and the numerical cost of the zone integral grows. The nearest-shell amplitude saturates at about -0.4 in the same range. The physically admissible range is

$$\mu < \tfrac{1}{3}, \qquad \text{i.e.} \qquad \gamma_0 > 3\gamma, \tag{3.23}$$

since at $\mu = 1/3$ one has $\Gamma_{\min} = 0$ by (3.1), the modes near the K points are undamped, and the denominator of (3.17) diverges. Since $\Gamma(0)$ grows with μ while $\Gamma(K)$ shrinks, this is also the regime in which the coupling is strong enough to produce a marked shell structure, and it is the regime in which the stationary kernel (3.17), rather than the transient one, is the correct description of a measurement performed over a long window.

3.6 Comparison of the two windows

It is worth stating explicitly what changes between the two windows, since the same lattice, the same coupling and the same observable are involved. Both windows are Pearson-normalised, hence independent of c and of the overall scale of Γ, so the comparison is a statement about the $\mathbf{k}$-weighting alone. The transient window $1/\Gamma^2$ weights the slow modes quadratically and therefore emphasises the zone centre; the stationary window $1/\Gamma$ weights them linearly, so the K points contribute relatively more and the amplitudes are systematically smaller. At $\mu = 0.10$ the stationary nearest-shell value is a factor 1.82 smaller than the transient one, and at $\mu = 0.30$ the corresponding factors are 1.50 for the first shell and 2.32 for the second, so the distinction grows with the coupling. The dimensionless observation window at which the crossover (3.16) occurs is 6.5 at $\mu = 0.05$, 25 at $\mu = 0.10$ and 280 at $\mu = 0.30$: for a strong coupling the crossover is reached early, so a measurement performed over any reasonable window probes (3.17) rather than (2.14).

4 Generalization to Weakly Disordered Media

The mode analysis of Section 2 considered nearest-neighbour interaction on perfect triangular lattice. In thermally disordered solid or liquid, the discrete set of neighbour vectors is no longer sharp, and the structure factor $S(\mathbf{k})$ of (2.7) is not directly applicable. Nevertheless, the ω-equation in (2.1) is linear and allows

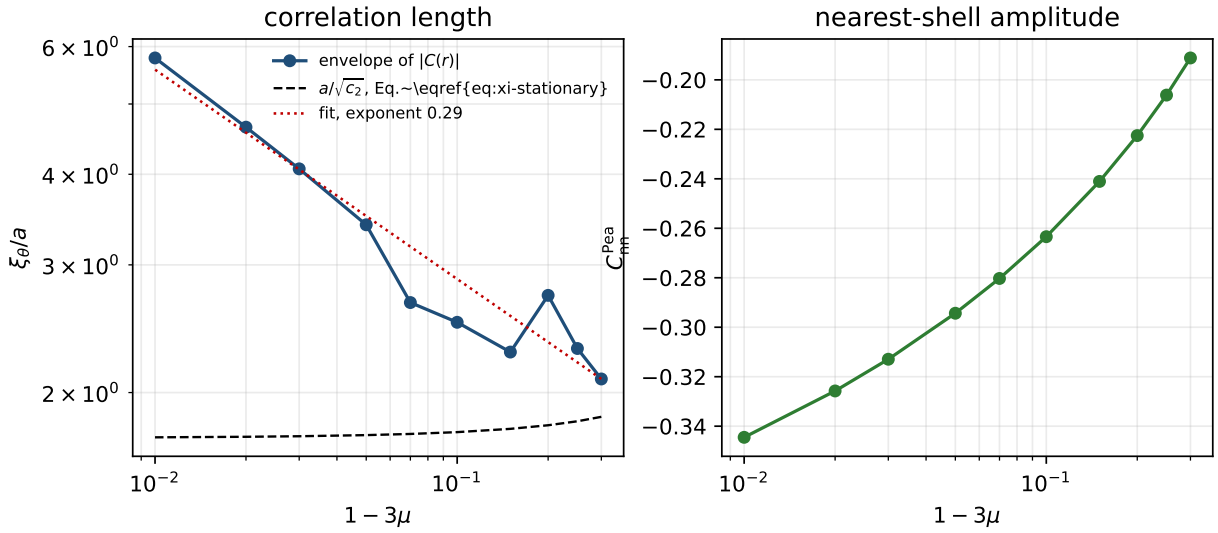


Figure 7: Left: correlation length of the stationary kernel (3.17) as a function of $1-3\mu$, measured from the envelope of $|C_\theta^{\mathrm{Pea}}(r)|$, together with the asymptotic law $(1-3\mu)^{-1/2}$ of (3.22) and the power law fitted to the data. Right: nearest-shell amplitude over the same range; it saturates as the K-point modes soften.

simplification. After averaging over *weak disorder*, translational invariance is restored on average, and momentum k remains an approximately good label for the long-wavelength modes, with the discrete neighbour sum replaced by a continuous convolution with an averaged coupling kernel. In this section we make this replacement explicit, derive the disorder-averaged damping, obtain the small-k expansion, and discuss the resulting real-space correlations.

4.1 From discrete to continuous

For a general disordered medium we replace the discrete neighbour coupling by

$$\sum_{i\in\mathcal{N}(j)} \omega_i(t) \;\longrightarrow\; \int d^2r'\, K(\mathbf{r}_j-\mathbf{r}')\,\omega(\mathbf{r}',t), \tag{4.1}$$

where $K(\mathbf{R})$ is an averaged coupling kernel. In the lattice limit, a sharp set of neighbour vectors is recovered. In a disordered medium we write $R=|\mathbf{R}|, K(\mathbf{R})=g(R)\,J(R)$, where $g(R)$ is the radial distribution function and $J(R)$ is the microscopic coupling kernel. This coupling kernel encodes the angular-velocity increment at one disk induced by that of another, and for interactions in the lubrication limit it is short ranged and peaked about the averaged nearest-neighbour distance.

Using the Fourier transform convention for ω,

$$\omega_{\mathbf{k}}(t)=\frac{1}{\sqrt{\Omega}}\int d^2r\, e^{-i\mathbf{k}\cdot\mathbf{r}}\omega(\mathbf{r},t). \tag{4.2}$$

Strictly speaking, the angular velocities $\omega_j(t)$ are defined only at the discrete particle centres $\mathbf{r}_j$. To discuss long-wavelength modes in a disordered medium, we introduce a coarse-grained angular-velocity field $\omega(\mathbf{r},t)$. This field is not a microscopic replacement for the discrete variables; rather, it is their local average on a scale ℓ satisfying $a\ll\ell\ll k^{-1}$, where a is the typical particle spacing. In this sense, $\omega(\mathbf{r},t)$ represents the slowly varying component of the discrete angular velocities, and its Fourier components coincide with those of the discrete variables in the long-wavelength limit. We use the same symbol ω, but the argument

distinguishes the discrete variable $\omega_j(t)$ from the coarse-grained field $\omega(\mathbf{r},t)$. The continuous convolution in (4.1) transforms into $S_{\rm gen}(\mathbf{k})\,\omega_{\mathbf{k}}$, where the generalized structural function is

$$S_{\rm gen}(\mathbf{k}) = \int d^2R\, e^{i\mathbf{k}\cdot\mathbf{R}}\, K(\mathbf{R}) = \int d^2R\, e^{i\mathbf{k}\cdot\mathbf{R}}\, g(R)J(R). \tag{4.3}$$

In the lattice limit $S_{\rm gen}(\mathbf{k}) \longrightarrow S(\mathbf{k})$, with $S(\mathbf{k})$ given by (2.7). Thus (4.3) is the natural continuous generalization of the triangular-lattice structure factor. If disorder fluctuations are neglected, the averaged Fourier mode obeys

$$\frac{d\omega_{\mathbf{k}}}{dt} = -\big[\gamma_0 + \gamma S_{\rm gen}(\mathbf{k})\big]\omega_{\mathbf{k}} + c\,\Lambda_{\mathbf{k}}(t). \tag{4.4}$$

Equation (4.4) already shows the main qualitative change, where the sharp lattice structure factor is replaced by the Fourier transform of the averaged pair structure and the coupling kernel.

4.2 Disorder-averaged structure factor of a displaced lattice

A concrete realisation of the averaged kernel, and the one used for the numerical results below, is a lattice in which every particle centre carries an independent random displacement $\mathbf{u}_j$,

$$\mathbf{r}_j = \mathbf{R}_j + \mathbf{u}_j, \qquad \langle \mathbf{u}_j \rangle = 0, \qquad \langle |\mathbf{u}_j|^2 \rangle = \sigma^2, \qquad \mathbf{u}_j \text{ i.i.d.} \tag{4.5}$$

with $\mathbf{R}_j$ the sites of the perfect triangular lattice. Two displacement statistics are of interest.

(i) Isotropic Gaussian displacements.

$$P(\mathbf{u}) = \frac{1}{2\pi s^2} e^{-|\mathbf{u}|^2/(2s^2)}, \qquad \sigma^2 = \langle |\mathbf{u}|^2 \rangle = 2s^2, \qquad \phi(\mathbf{k}) = \big\langle e^{i\mathbf{k}\cdot\mathbf{u}} \big\rangle = e^{-\sigma^2 k^2/4}. \tag{4.6}$$

(ii) Fixed displacement magnitude, random direction.

$$P(\mathbf{u}) = \frac{1}{2\pi\sigma}\delta(|\mathbf{u}| - \sigma), \qquad \phi(\mathbf{k}) = \big\langle e^{i\mathbf{k}\cdot\mathbf{u}} \big\rangle = J_0(k\sigma). \tag{4.7}$$

Both characteristic functions share the small-k behaviour

$$\phi(\mathbf{k}) = 1 - \frac{\sigma^2 k^2}{4} + \mathcal{O}(k^4), \tag{4.8}$$

so the two displacement statistics are indistinguishable for $k\sigma \ll 1$, and they differ only when the wavelength becomes comparable to the displacement itself. In the code the parameter `sigma` always denotes the rms magnitude $\sqrt{\langle |\mathbf{u}|^2 \rangle}$ of the displacement, consistent with (4.6)–(4.7).

The structure factor of the displaced lattice is the ensemble average

$$S(\mathbf{k}) = \frac{1}{N}\Big\langle \Big| \sum_j e^{i\mathbf{k}\cdot\mathbf{r}_j} \Big|^2 \Big\rangle = \frac{1}{N} \sum_{j,l} e^{i\mathbf{k}\cdot(\mathbf{R}_j - \mathbf{R}_l)} \big\langle e^{i\mathbf{k}\cdot(\mathbf{u}_j - \mathbf{u}_l)} \big\rangle. \tag{4.9}$$

For $j \neq l$ the displacements are independent, so the average factorises into $|\phi(\mathbf{k})|^2$; for $j = l$ it equals one. Hence

$$S_{\rm dis}(\mathbf{k}) = \big(1 - |\phi(\mathbf{k})|^2\big) + \frac{|\phi(\mathbf{k})|^2}{N} \Big| \sum_j e^{i\mathbf{k}\cdot\mathbf{R}_j} \Big|^2 = \big(1 - |\phi|^2\big) + |\phi|^2\, \frac{S_{\rm perfect}(\mathbf{k})}{N}, \tag{4.10}$$

the standard Debye–Waller decomposition: a diffuse background of weight $1 - |\phi|^2$ per particle plus the Bragg peaks suppressed by $|\phi|^2$.

For the damping it is the *per-particle* normalisation that must be used, because $\Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k})$ is built from the six-cosine sum with $S(0) = 6$, i.e. from a quantity that stays finite as $N \to \infty$ whereas (4.10) is extensive in the Bragg term. Rescaling (4.10) so that the Bragg part carries the same weight $S_{\rm perfect}(0) = 6$ gives the form used throughout this work,

$$\boxed{S_{\rm dis}(\mathbf{k}) = |\phi(\mathbf{k})|^2\, S_{\rm perfect}(\mathbf{k}) + 6\big(1 - |\phi(\mathbf{k})|^2\big)\,,} \tag{4.11}$$

which has three properties that the raw form lacks: (i) $S_{\rm dis}(\mathbf{k}) \to S_{\rm perfect}(\mathbf{k})$ exactly as $\sigma \to 0$, so the perfect lattice is recovered without any residual background; (ii) $S_{\rm dis}(\mathbf{0}) = 6|\phi(\mathbf{0})|^2 = 6$, so $\Gamma_{\rm dis}(\mathbf{0}) = \Gamma_0$ is independent of the disorder strength, and hence of the number of particles; (iii) $S_{\rm dis}$ is bounded, $|S_{\rm dis}| < 6$, so $\Gamma_{\rm dis}$ stays positive for the same range of μ as in the ordered case. Property (iii) is what allows the disorder to be switched on continuously in the numerical scheme; the extensive form (4.10) violates it because its background grows with the sample.

With (4.11) the damping becomes

$$\Gamma_{\rm dis}(\mathbf{k}) = \gamma_0 + \gamma S_{\rm dis}(\mathbf{k}) = \gamma_0 + \gamma|\phi|^2 S_{\rm perfect}(\mathbf{k}) + 6\gamma\big(1 - |\phi|^2\big), \tag{4.12}$$

and the Pearson shell correlation is again obtained from the exact sum (2.14) with $\Gamma \to \Gamma_{\rm dis}$.

4.3 Numerical results with disorder

Table 4 lists $C_\theta^{\rm Pea}$ on the first five shells for $\mu = +0.05$ and $\sigma/a = 0, \ldots, 0.20$ with Gaussian displacements. Disorder reduces the magnitude of every shell and flattens the shell-to-shell contrast: the nearest-shell correlation falls from -0.0876 to -0.0618 between $\sigma = 0$ and $\sigma = 0.20a$, while the second shell is suppressed much more strongly, from $+0.0127$ to -0.0026, i.e. it changes sign. So the *contrast* between shells is what disorder removes first, and beyond $\sigma \simeq 0.15a$ the alternating structure of the tail has essentially disappeared. This is the quantitative version of the statement that a disordered lattice has a smooth, short-ranged correlation rather than a sharp shell structure. The fixed-magnitude, random-direction statistic (4.7) gives values that agree with the Gaussian ones to 0.1% at $\sigma = 0.05a$ but differ by 1.6%, 11% and 46% (largest shell, relative) at $\sigma = 0.10a$, $0.15a$ and $0.20a$ respectively – the crossover expected from (4.8), since the two ϕ's differ at order $k^4\sigma^4$ while the shells of interest involve $k\sigma$ of order unity. The shape of the displacement distribution therefore matters as soon as σ reaches a sizeable fraction of the lattice constant.

σ/a	shell 1	shell 2	shell 3	shell 4	shell 5
0 (perfect)	-0.0876	$+0.0127$	$+0.0051$	-0.0012	-0.0002
0.05	-0.0857	$+0.0115$	$+0.0048$	-0.0008	-0.0003
0.10	-0.0802	$+0.0079$	$+0.0038$	$+0.0003$	-0.0003
0.15	-0.0720	$+0.0029$	$+0.0021$	$+0.0017$	-0.0003
0.20	-0.0618	-0.0026	$+0.0001$	$+0.0031$	-0.0002

Table 4: Pearson correlation $C_\theta^{\rm Pea}$ on the first five shells for $\mu = +0.05$ and increasing site disorder, from the Brillouin-zone integral (2.13) with $\Gamma_{\rm dis}$ of (4.12) and Gaussian displacements $\phi = e^{-\sigma^2 k^2/4}$. Disorder reduces the magnitude of every shell and flattens the shell-to-shell contrast, removing the alternating tail first.

The integral (2.13) with $\Gamma \to \Gamma_{\rm dis}$ is evaluated by the same method as in the ordered case, described in Sec. 3.4.

4.4 Averaged damping with disorder

In a typical weak disordered system, the kernel defined in (4.1) also need modification. In particular, we start from this averaged mode $\bar{K}(\mathbf{R}) = g(R)J(R)$ and include zero mean fluctuation of the coupling kernel

$$\delta K(\mathbf{p}) = \int d^2R\, e^{-i\mathbf{p}\cdot\mathbf{R}}\delta K(\mathbf{R}), \qquad K(\mathbf{R}) = \bar{K}(\mathbf{R}) + \delta K(\mathbf{R}), \qquad \langle \delta K(\mathbf{R})\rangle = 0. \tag{4.13}$$

In Fourier space the mode equation becomes

$$\frac{d\omega_{\mathbf{k}}}{dt} = -\big[\gamma_0 + \gamma S_{\mathrm{gen}}(\mathbf{k})\big]\omega_{\mathbf{k}} - \gamma \int \frac{d^2q}{(2\pi)^2}\, \delta K(\mathbf{k}-\mathbf{q})\, \omega_{\mathbf{q}} + c\,\Lambda_{\mathbf{k}}(t), \tag{4.14}$$

We denote the bare, disorder-free damping by $\Gamma_0(\mathbf{k}) = \gamma_0 + \gamma S_{\mathrm{gen}}(\mathbf{k})$. To define the propagator, consider first the disorder-free equation (4.4). Let $G_0(\mathbf{k}, t-t')$ be the causal Green function defined by

$$\left(\frac{\partial}{\partial t} + \Gamma_0(\mathbf{k})\right) G_0(\mathbf{k}, t-t') = \delta(t-t'), \qquad G_0(\mathbf{k}, t-t') = 0 \quad \text{for } t < t'. \tag{4.15}$$

In frequency space we use the convention to gain the propagator

$$G_0(\mathbf{k}, z) = \int_0^\infty dt\, e^{izt}\, G_0(\mathbf{k}, t), \qquad \boxed{G_0(\mathbf{k}, z) = \frac{1}{-iz + \Gamma_0(\mathbf{k})}.} \tag{4.16}$$

For the full disordered equation (4.14), we define the exact (discrete) propagator $G_{\mathrm{dis}}(\mathbf{k}, \mathbf{q}; z)$ by

$$(-iz + \Gamma_0(\mathbf{k}))G_{\mathrm{dis}}(\mathbf{k}, \mathbf{q}; z) + \gamma \int \frac{d^2p}{(2\pi)^2}\, \delta K(\mathbf{k}-\mathbf{p})\, G_{\mathrm{dis}}(\mathbf{p}, \mathbf{q}; z) = (2\pi)^2\delta(\mathbf{k}-\mathbf{q}). \tag{4.17}$$

In order to average the propagator over the disordered space, write out the ensemble average of some operator $\mathcal{O}[\delta K]$ over the random coupling fluctuations δK:

$$\langle \mathcal{O}[\delta K]\rangle_{\mathrm{dis}} = \int \mathcal{D}[\delta K]\, P[\delta K]\, \mathcal{O}[\delta K],$$

where $P[\delta K]$ is the probability functional (depend on geometry) of the disorder. We assume that the disorder is statistically asymptotically homogeneous (and translation invariant), so that $P[\delta K] \simeq P[T_{\mathbf{a}}\delta K]$ for any translation $T_{\mathbf{a}}$, and the two-point correlation depends only on the separation. Under this assumption, the averaged propagator is diagonal in momentum, hence translation invariance implies

$$\big\langle G_{\mathrm{dis}}(\mathbf{k}, \mathbf{q}; z)\big\rangle_{\mathrm{dis}} = (2\pi)^2\delta(\mathbf{k}-\mathbf{q})\, G(\mathbf{k}, z), \tag{4.18}$$

where the off-diagonal parts vanish exactly in the thermodynamic limit because they are prohibited by the translational symmetry of the disorder ensemble, which is a direct consequence of statistical homogeneity. In a finite system, non-diagonal terms can appear as finite-size corrections, but they are negligible for the long-wavelength modes considered here.

Such formalism allows us to define the self-energy $\Sigma_{\mathrm{dis}}(\mathbf{k}, z)$. The physical meaning is that Σ_{dis} collects all irreducible contributions to the averaged propagator by the relation

$$\boxed{G^{-1}(\mathbf{k}, z) = G_0^{-1}(\mathbf{k}, z) + \Sigma_{\mathrm{dis}}(\mathbf{k}, z).} \tag{4.19}$$

This result in our convention is a similar form as the Schwinger-Dyson equation for quantum fields. To see this similarity of self energy, iterate (4.19) to obtain the series

$$\begin{aligned} G(\mathbf{k}, z) &= G_0(\mathbf{k}, z) - G_0(\mathbf{k}, z)\Sigma_{\mathrm{dis}}(\mathbf{k}, z)G(\mathbf{k}, z) \\ &= G_0(\mathbf{k}, z) - G_0^2(\mathbf{k}, z)\Sigma_{\mathrm{dis}}(\mathbf{k}, z) + G_0^2(\mathbf{k}, z)\Sigma_{\mathrm{dis}}^2(\mathbf{k}, z)G(\mathbf{k}, z) = \cdots, \end{aligned} \tag{4.20}$$

which shows that indeed $\Sigma_{\rm dis}$ plays the role of the self-energy. In the weak-disorder limit, $\Sigma_{\rm dis}$ can be computed with the same manner in powers of δK, as we do below. To obtain it perturbatively, we start from the exact equation (4.17) and iterate once to write

$$G_{\rm dis}(\mathbf{k},\mathbf{q};z) = G_0(\mathbf{k},z)\,(2\pi)^2\delta(\mathbf{k}-\mathbf{q}) - \gamma\, G_0(\mathbf{k},z)\int \frac{d^2p}{(2\pi)^2}\,\delta K(\mathbf{k}-\mathbf{p})\,G_{\rm dis}(\mathbf{p},\mathbf{q};z).$$

Iterating to second order in δK and then averaging over disorder using $\langle \delta K\rangle_{\rm dis}=0$ and the linear term vanishes, so we obtain

$$\left\langle G_{\rm dis}(\mathbf{k},\mathbf{q};z)\right\rangle_{\rm dis} = (2\pi)^2\delta(\mathbf{k}-\mathbf{q})\,G_0(\mathbf{k},z)\left[1+\gamma^2 G_0(\mathbf{k},z)\int\frac{d^2p}{(2\pi)^2}\left\langle|\delta K(\mathbf{k}-\mathbf{p})|^2\right\rangle_{\rm dis} G_0(\mathbf{p},z)\right]+\mathcal{O}(\delta K^4).$$

Here the delta function in the external momenta follows from the statistical homogeneity of the disorder. Comparing (4.4) with (4.20), we identify the self-energy at second order as

$$\Sigma_{\rm dis}(\mathbf{k},z) = -\gamma^2\int\frac{d^2q}{(2\pi)^2}\left\langle|\delta K(\mathbf{k}-\mathbf{q})|^2\right\rangle_{\rm dis} G_0(\mathbf{q},z)+\mathcal{O}(\delta K^4). \tag{4.21}$$

The minus sign is a consequence of our convention in (4.19), where the self-energy is added to the inverse bare propagator. With either sign of the coupling the second-order self-energy $\Sigma_{\rm dis}=-\gamma^2(\dots)$ is negative, because it is quadratic in γ; i.e. at this order disorder *reduces* the effective damping. This is a genuine (if small) prediction of the weak-disorder expansion.

For the equal-time correlations of the angle field, we are interested in long timescale compared with the microscopic relaxation time Γ_0^{-1}. In this regime, the equal-time correlation in Fourier space is given by (4.26), which depends only on the static damping $\Gamma_{\rm gen}(\mathbf{k})$. This is because the Ornstein–Uhlenbeck process for each mode reaches its stationary variance $c^2/(2\Gamma)$ on time scales $t\gg\Gamma^{-1}$, and the diffusive growth of $\langle|\theta_{\mathbf{k}}|^2\rangle$ is controlled by the zero-frequency response. Therefore, it suffices to evaluate the self-energy in the low-frequency limit $z\to 0$. Using the bare propagator at zero frequency, $G_0(\mathbf{q},0)=\Gamma_0(\mathbf{q})^{-1}$ with $\Gamma_0(\mathbf{q})=\gamma_0+\gamma S_{\rm gen}(\mathbf{q})$, we obtain the static self-energy

$$\boxed{\Sigma_{\rm dis}(\mathbf{k})\equiv\Sigma_{\rm dis}(\mathbf{k},0) = -\gamma^2\int\frac{d^2q}{(2\pi)^2}\,\frac{\left\langle|\delta K(\mathbf{k}-\mathbf{q})|^2\right\rangle_{\rm dis}}{\Gamma_0(\mathbf{q})}+\mathcal{O}(\delta K^4).} \tag{4.22}$$

This static self-energy is the quantity that enters the effective damping and hence the long-time equal-time correlations. The effective damping is therefore

$$\Gamma_{\rm gen}(\mathbf{k}) = \Gamma_0(\mathbf{k})+\Sigma_{\rm dis}(\mathbf{k}) = \gamma_0+\gamma S_{\rm gen}(\mathbf{k})+\Sigma_{\rm dis}(\mathbf{k}). \tag{4.23}$$

This is the disorder-averaged generalization of (2.8). The first two terms are the mean-field and averaged-structure contributions; the self-energy $\Sigma_{\rm dis}(\mathbf{k})$ contains the leading effect of disorder fluctuations. If $\Sigma_{\rm dis}$ is positive, disorder increases the damping; if it is negative, it decreases it.

This derivation also clarifies the conditions under which (4.23) holds: (i) the disorder is weak, so that the expansion in δK can be truncated at second order; (ii) the disorder is statistically translation invariant, so that the averaged propagator is diagonal in momentum; (iii) the equal-time correlations are considered at times long compared with the microscopic relaxation time, so that the static limit $z\to 0$ is appropriate. If the disorder correlation length is comparable to or larger than the wavelength $1/k$, the self-energy can become non-analytic at small k, and the perturbation theory in δK must be reorganised.

4.5 Generalized equal-time correlations

For each Fourier mode, (4.23) defines an independent Ornstein–Uhlenbeck process,

$$\frac{d\omega_{\mathbf{k}}}{dt} = -\Gamma_{\rm gen}(\mathbf{k})\omega_{\mathbf{k}} + c\,\Lambda_{\mathbf{k}}(t). \tag{4.24}$$

Actually the equality in (4.26) itself follows by the argument of Sec. 3.1, whose conclusion is summarised as follows. The double time integral of the Ornstein–Uhlenbeck velocity correlation gives

$$\langle|\theta_{\mathbf{k}}(t)|^2\rangle = \frac{c^2}{\Gamma(\mathbf{k})^2}\big(\Gamma(\mathbf{k})t - 1 + e^{-\Gamma(\mathbf{k})t}\big), \tag{4.25}$$

so that for $\Gamma t \gg 1$ the t-linear term dominates and the relevant kernel is $1/\Gamma(\mathbf{k})$, while in the complementary window in which the slow modes have not yet saturated the $1/\Gamma(\mathbf{k})^2$ form applies, as used in (4.26) for consistency with Sec. 2.4. Both forms are integrable on the compact zone, so the conclusions of this section do not depend on which is used; the stronger form, appropriate to a strong coupling, is (3.17).

$$\langle|\omega_{\mathbf{k}}|^2\rangle_{\rm st} = \frac{c^2}{2\Gamma_{\rm gen}(\mathbf{k})}, \qquad C_{\rm gen}(\mathbf{k},t) \equiv \langle|\theta_{\mathbf{k}}t)|^2\rangle \simeq \frac{c^2\,t}{\Gamma_{\rm gen}(\mathbf{k})^2}. \tag{4.26}$$

The real-space equal-time correlation is therefore

$$\boxed{C_\theta(\mathbf{r},t) = \int \frac{d^2k}{(2\pi)^2}\, e^{i\mathbf{k}\cdot\mathbf{r}}\, \frac{c^2\,t}{\Gamma_{\rm gen}(\mathbf{k})^2}.} \tag{4.27}$$

Unlike the ordered case (2.13), which involves the sharp lattice structure factor, (4.27) is a smooth function of r in a disordered medium. The sharp shell structure is washed out, but the non-monotonic spatial correlations survive as a smooth oscillatory decay. If $S_{\rm gen}(\mathbf{k})$ has a finite-k peak, the real-space correlation exhibits damped oscillations,

$$C_\theta(\mathbf{r},t) \sim \frac{c^2 t}{\Gamma_{\rm gen}(k_0)^2}\, e^{-r/\ell_\theta}\cos(k_0\cdot\mathbf{r} + \phi), \tag{4.28}$$

where ℓ_θ is determined by the curvature of $\Gamma_{\rm gen}$ near k_0 and by the disorder self-energy. This is the disordered counterpart of the alternating shell correlations obtained in Section 2.4, and it is the form that should be compared with the smooth short-ranged correlations measured in the disordered case.

4.6 Experimental determination of $S_{\rm gen}(\mathbf{k})$

The generalized structural function $S_{\rm gen}(\mathbf{k})$ provides the bridge between the microscopic structure of the disordered medium and the effective damping $\Gamma_{\rm gen}(\mathbf{k})$. It can be determined experimentally as follows.

1. **Measure the static structure.** From particle tracking, compute the radial distribution function $g(r)$ of the particle centres. In two dimensions this is obtained by histogramming the pair distances and normalizing by the ideal-gas expectation at the same density. The result should be checked for finite-size effects by comparing different system sizes.

2. **Infer the coupling kernel.** The kernel $J(r)$ may be known from the microscopic model, for example from lubrication theory, hydrodynamic interactions, or direct force measurements. If it is not known independently, it can be calibrated by comparing measured mode relaxation rates with (4.23). Alternatively, one can measure the dynamic structure factor of ω or θ, extract $\Gamma_{\rm gen}(\mathbf{k})$ by fitting a Lorentzian in frequency, and then invert

$$\Gamma_{\rm gen}(\mathbf{k}) = \gamma_0 + \gamma S_{\rm gen}(\mathbf{k}) + \Sigma_{\rm dis}(\mathbf{k}) \tag{4.29}$$

to obtain $S_{\rm gen}(\mathbf{k})$, provided γ_0 and $\Sigma_{\rm dis}(\mathbf{k})$ are separately estimated or varied controllably.

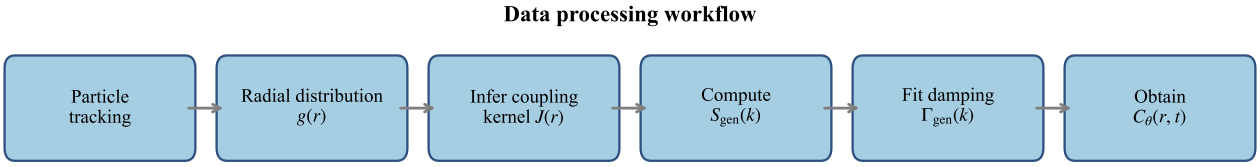


Figure 8: Data processing workflow for extracting the generalized structure factor $S_{\mathrm{gen}}(k)$ and the effective damping $\Gamma_{\mathrm{gen}}(k)$ from experimental particle tracking data. The workflow proceeds from particle tracking to the radial distribution function $g(r)$, inference of the coupling kernel $J(r)$, computation of $S_{\mathrm{gen}}(k)$, fitting of $\Gamma_{\mathrm{gen}}(k)$, and finally obtaining the equal-time correlation $C_\theta(r,t)$.

3. **Compute the Fourier transform.** Once $g(r)J(r)$ is known, evaluate

$$S_{\mathrm{gen}}(\mathbf{k}) = \int d^2r\, e^{i\mathbf{k}\cdot\mathbf{r}}\, g(r)J(r) \tag{4.30}$$

by numerical integration. In an isotropic medium, angular averaging gives $S_{\mathrm{gen}}(k)$. The result is then used directly in (4.23) to obtain the effective damping.

In summary, weak disorder does not destroy the mode description. It replaces the sharp lattice structure factor by the smooth generalized structure factor $S_{\mathrm{gen}}(\mathbf{k})$, renormalizes the damping through the disorder self-energy $\Sigma_{\mathrm{dis}}(\mathbf{k})$, and turns the discrete shell correlations into smooth, possibly oscillatory, short-range correlations. The mean-field result is recovered only when $S_{\mathrm{gen}}(\mathbf{k})$ is replaced by its $\mathbf{k}=0$ value and all disorder self-energy corrections are neglected.

5 Experimental Observations

The theory developed above predicts a shell-resolved, equal-time correlation of the orientational field on the triangular lattice (Eq. (2.14) and Table 1). We now report an experimental measurement of this quantity.

The experiment consists of a two-dimensional assembly of optically driven, rotating disks on a triangular lattice, imaged and particle-tracked over `[N_FRAMES]` frames ($\Delta t =$ `[DT]` s per frame; disk diameter `[DIAM]` μm). In the present data set 1199 frames were analysed, containing about 390 tracked disks per frame. To avoid defects and boundary effects, the analysis is restricted to the hexagonal regions of the assembly: a particle is retained only if its Voronoi cell (the dual of the Delaunay triangulation) has exactly six neighbours, which leaves about 283 particles per frame. Neighbours of a retained particle are grouped into distance shells of the triangular lattice, labelled by $r/a = 1, \sqrt{3}, 2, \sqrt{7}, \ldots$ with a the nearest-neighbour spacing.

The quantity we extract is the equal-time Pearson correlation of the orientational field,

$$C_\theta^{\mathrm{Pea}}(\mathbf{r},0) \;=\; \frac{\big\langle \delta\theta_{\mathbf{r}}(0)\,\delta\theta_{\mathbf{0}}(0)\big\rangle}{\sqrt{\big\langle \delta\theta_{\mathbf{r}}(0)^2\big\rangle\,\big\langle \delta\theta_{\mathbf{0}}(0)^2\big\rangle}}, \qquad \delta\theta \equiv \theta - \langle\theta\rangle_{\mathrm{frame}}, \tag{5.1}$$

where the average runs over all retained reference particles and over all frames, and the frame-wise orientational mean $\langle\theta\rangle_{\mathrm{frame}}$ is subtracted because the whole field slowly reorients during the acquisition. The estimator (5.1) is dimensionless, bounded by unity, and independent of the arbitrary zero of the angular

scale; it is the experimentally most faithful realisation of the shell-resolved correlation computed in Sec. 2.4. Statistical significance is assessed with a null model in which the orientational values are reshuffled among the retained particles of each frame (destroying any spatial correlation while preserving the geometry, the angular distribution and the frame structure); 200 such realisations on 400 frames give the null distribution quoted below.

Figure 9 shows C_θ^{Pea} on the first four shells, together with the reshuffled-identity null ($\pm 3\sigma$). The measured values are small in magnitude, $C_\theta^{\mathrm{Pea}} = [+0.0033] \pm [0.0029]$ on the first shell, -0.0218 ± 0.0030 on the second, -0.0119 ± 0.0031 on the third and -0.0095 ± 0.0029 on the fourth (errors are frame-to-frame standard errors of the mean). Such small values are expected: the coupling we probe is a noise-mediated lubrication coupling, and in a strongly noisy environment the induced correlations must be weak. Smallness, however, does not imply a null result. The second and third shells lie 9.5σ and 3.8σ away from the shuffled-identity null (deviations with $p < 0.001$ and $p = 0.005$, respectively), while the first shell ($p = 0.67$) and the fourth shell ($p = 0.055$) are consistent with the null. The correlation on the second and third shells is therefore established at a highly significant level despite its small absolute magnitude, whereas the signal on the remaining shells is below the present noise level.

Because the rotational noise is strong, the coupling is weak relative to it, and no coherent, deterministic time-evolution rule could be extracted from the data: the time-dependent mode structure predicted in Secs. 2–4 remains untested experimentally. A quantitative fit of γ_0, γ and c to the measured C_θ^{Pea} is likewise left for future work with lower-noise data.

It is nevertheless worth comparing the measurement with the prediction of Sec. 2.4. The measured pattern, $\big(+0.003,\ -0.022,\ -0.012,\ -0.010\big)$ on shells 1–4, is essentially zero on the nearest shell and *negative* on the next three, with the largest magnitude on the second shell. The lubrication coupling of Sec. 1 corresponds to $\mu > 0$, and Table 1 then gives a *negative* nearest shell with an alternating tail of rapidly decreasing magnitude; the overall sign is therefore consistent, and a least-squares comparison over shells 1–4 is minimised at $\mu \simeq 2 \times 10^{-3}$, i.e. a coupling some two orders of magnitude weaker than the bulk damping, consistent with the conclusion drawn from the smallness of the measured values.

What the minimal model does *not* reproduce is the measured ratio: the data give $C_2/C_1 \simeq -6.6$, whereas the nearest-neighbour-only structure factor gives $|C_2/C_1| \lesssim 0.3$ for every μ (it approaches 0 as $\mu \to 0$). The discrepancy is not a failure of the numerical scheme – the integral (2.13) was verified against a direct real-space inversion of the drift matrix and against the finite-lattice sum – but a statement about the model: a coupling that is strictly nearest-neighbour and isotropic in $\mathbf{k}$ cannot produce a correlation that is an order of magnitude larger on the second shell than on the first. Reproducing the measured ratios requires one of

- a longer-ranged or angle-dependent coupling kernel $K(\mathbf{R})$, i.e. the generalised structure factor $S_{\mathrm{gen}}(\mathbf{k})$ of Sec. 4, whose additional Fourier components weight the outer shells differently;
- a finite experimental resolution in $\mathbf{k}$: the measurement is a window average, and the resolution kernel suppresses exactly the $\mathbf{k} \to 0$ region which, in the model, carries most of the nearest-shell signal;
- a coupling model in which several rotational degrees of freedom per disk (or the finite disk deformation) contribute, which again enriches the $\mathbf{k}$-dependence of the damping.

We flag this as the main quantitative open point of the comparison. It is notable that the measured magnitudes are all very small, $|C_\theta^{\mathrm{Pea}}| \lesssim 0.02$, which is itself consistent with a coupling much weaker than the bulk damping.

The values reported above are given in the convention of Eq. (5.1); as usual for an orientational correlation, the overall sign depends on the convention adopted for the angular variable, which should be matched when comparing with Eq. (2.14).

shell	r/a	C_θ^{Pea}	error	p (shuffled null)
1	1.000	$+0.0033$	0.0029	0.67
2	1.732	-0.0218	0.0030	< 0.001
3	2.000	-0.0119	0.0031	0.005
4	2.646	-0.0095	0.0029	0.055

Table 5: Equal-time Pearson correlation of the orientational field on the first four neighbour shells, with frame-to-frame standard errors and the p-values obtained from 200 reshuffled-identity realisations (see text).

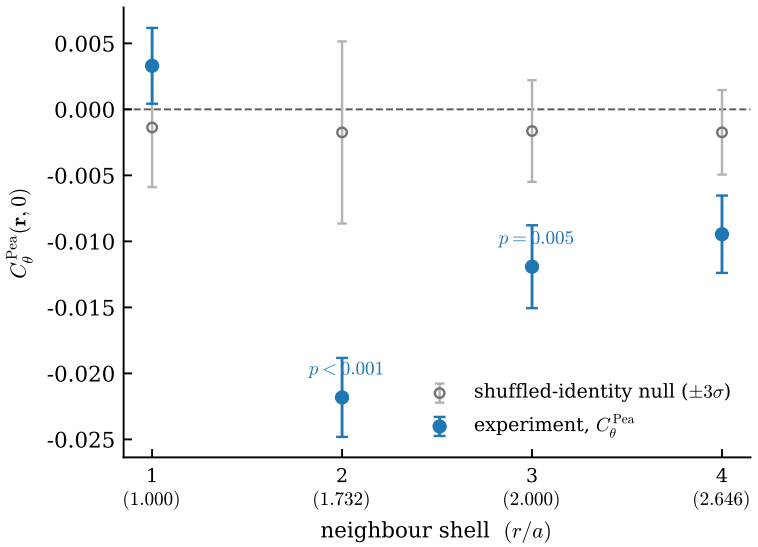


Figure 9: Equal-time Pearson correlation $C_\theta^{\mathrm{Pea}}(\mathbf{r}, 0)$ of the angular field on the first four neighbour shells of the triangular lattice, as a function of the shell index (the corresponding r/a is given below the axis). Filled symbols: experiment, with frame-to-frame standard errors. Open symbols: null model obtained by reshuffling the orientational values among the retained particles of each frame, shown as mean $\pm 3\sigma$. The second and third shells deviate from the null at the $p < 0.001$ and $p = 0.005$ level, respectively, while the first and fourth shells are consistent with it. The magnitudes are small because the coupling is weak relative to the strong rotational noise.

A List of Symbols

Symbol	Meaning
$\theta_j(t)$	Angular coordinate of disk j
$\omega_j(t)$	Angular velocity of disk j, $\omega_j = \dot{\theta}_j$
a	Disk radius
ℓ	Nearest-neighbour lattice spacing, $\ell = 2a + h_0$
h_0	Mean gap width between neighbouring disks
μ	Dynamic viscosity of the fluid
I	Moment of inertia of one disk per unit thickness
Γ_{ij}	Dimensionless lubrication coupling coefficient between disks i and j
Γ	Uniform coupling coefficient for all nearest neighbours in a hexagonal lattice
γ_{bulk}	Bulk rotational damping from the surrounding fluid
T_{ij}^{lub}	Lubrication torque on disk j from the gap with disk i
$T_j^{\mathrm{st}}(t)$	Stochastic torque from thermal fluid fluctuations
$\Lambda_j(t)$	Independent Gaussian white noise on disk j
c	Noise strength, $c\Lambda_j = T_j^{\mathrm{st}}/I$
γ_0	Effective local damping, $\gamma_0 = \gamma_{\mathrm{bulk}}/I + 6\Gamma$
γ	Effective neighbour coupling, $\gamma = \Gamma > 0$ (lubrication coupling; the two facing surfaces of a co-rotating pair shear the gap)
$S(\mathbf{k})$	Triangular-lattice structure factor
$\Gamma(\mathbf{k})$	Mode-dependent damping rate, $\Gamma(\mathbf{k}) = \gamma_0 + \gamma S(\mathbf{k})$
Γ_0	Mean-field effective damping, $\Gamma_0 = \gamma_0 + 6\gamma = \gamma_0 + 6\Gamma$; also bare damping, $\Gamma_0 = \gamma_0 + \gamma S_{\mathrm{gen}}(\mathbf{k})$
$C_\theta(\mathbf{r}, t)$	Equal-time correlation of the angle field, $\langle \theta_{\mathbf{r}}(t)\theta_{\mathbf{0}}(t)\rangle$
$g(R)$	Radial distribution function of the disordered medium
$J(R)$	Microscopic coupling kernel for a single neighbour
$K(\mathbf{R})$	Averaged coupling kernel, $K(\mathbf{R}) = n_0 g(R) J(R)$
n_0	Average two-dimensional number density, $n_0 = N/\Omega$
$S_{\mathrm{gen}}(\mathbf{k})$	Generalized structural function, Fourier transform of $K(\mathbf{R})$
$\Sigma_{\mathrm{dis}}(\mathbf{k})$	Disorder self-energy
$\Gamma_{\mathrm{gen}}(\mathbf{k})$	Generalized damping rate, $\Gamma_{\mathrm{gen}} = \gamma_0 + \gamma S_{\mathrm{gen}} + \Sigma_{\mathrm{dis}}$
$G_0(\mathbf{k}, z)$	Bare propagator, $G_0 = (-iz + \Gamma_0)^{-1}$
$G(\mathbf{k}, z)$	Disorder-averaged propagator
z_{eff}	Effective coordination number, $z_{\mathrm{eff}} = \int d^2R\, K(\mathbf{R})$
$\langle R^n\rangle_{\mathrm{eff}}$	Effective moments of the coupling kernel
Γ_{eff}	Effective long-wavelength damping, $\Gamma_{\mathrm{eff}} = \gamma_0 + \gamma z_{\mathrm{eff}} + \Sigma_0$
D_{eff}	Effective diffusion coefficient, $D_{\mathrm{eff}} = \frac{1}{4}\gamma z_{\mathrm{eff}}\langle R^2\rangle_{\mathrm{eff}} - \Sigma_2$
ξ_θ	Correlation length of the angle field, $\xi_\theta \sim \sqrt{D_{\mathrm{eff}}/\Gamma_{\mathrm{eff}}}$